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Hy-DeFake: Hypergraph neural networks for detecting fake news in online social networks

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ABSTRACT

Nowadays social media is the primary platform for people to obtain news and share information. Combating online fake news has become an urgent task to reduce the damage it causes to society. Existing methods typically improve their fake news detection performances by utilizing textual auxiliary information (such as relevant retweets and comments) or simple structural information (*i.e.*, graph construction). However, these methods face two challenges. First, an increasing number of users tend to directly forward the source news without adding comments, resulting in a lack of textual auxiliary information. Second, simple graphs are unable to extract complex relations beyond pairwise association in a social context. Given that real-world social networks are intricate and involve high-order relations, we argue that exploring beyond pairwise relations between news and users is crucial for fake news detection. Therefore, we propose constructing an attributed hypergraph to represent non-textual and high-order relations for user participation in news spreading. We also introduce a hypergraph neural network-based method called Hy-DeFake to tackle the challenges. Our proposed method captures semantic information from news content, credibility information from involved users, and high-order correlations between news and users to learn distinctive embeddings for fake news detection. The superiority of Hy-DeFake is demonstrated through experiments conducted on four widely-used datasets, and it is compared against nine baselines using four evaluation metrics.

1. Introduction

Nowadays, social media platforms such as Facebook and Twitter have emerged as primary sources for accessing timely news and sharing information. While these platforms offer the advantage of instant access to real-time news from diverse perspectives, the ease of engagement and interaction has led to the rapid proliferation of fake news. This spread of misinformation has resulted in significant physical and mental harm to individuals. For instance, over 5,800 people were admitted to hospital after falling victim to a piece of COVID-19 fake news, which claims that alcohol with high concentration can sanitize the body and eradicate the virus (Islam et al., 2020). Fake news not only poses a threat to public health but also undermines trust in governments and alters public perception (DiFranzo & Gloria-Garcia, 2017). For example, research has shown that fake news circulating on social media played a role in shaping the vote for the Brexit referendum (Bastos & Mercea, 2019). During the Brexit campaign, a significant amount of false information circulated on social platforms, with pro-Leave fake news spreading more widely on Twitter than pro-Remain fake news. The widespread dissemination of fake news has had detrimental social consequences. Therefore, it is an imminent task to identify fake

news online to ensure the public receives trustworthy information and reliable guidance.

Various methods have been developed for the detection of fake news in online social networks (Guo, Ding, Yao, Liang, & Yu, 2020). A simple technique to identify fake news is through text classification, but the results indicate that these approaches have limited performance since they only consider news content itself and overlook the social context (Pérez-Rosas, Kleinberg, Lefevre, & Mihalcea, 2018). To achieve more accurate results, some studies adopt textual data from social context as auxiliary information, such as comments, tweets, and retweets of source news (Ma, Gao, & Wong, 2018; Rao, Miao, Jiang, & Li, 2021; Shu, Cui, Wang, Lee, & Liu, 2019). However, these methods face a practical problem: many users tend to retweet the news directly without adding comments or expressing their stances, which results in the absence of textual side information. Let us take the spreading of a piece of COVID-19 news as an example, as shown in Fig. 1. Most users participate in sharing this news by directly retweeting it, such as #User1, #User7. The comments published during retweeting are often insignificant, like the symbols from #User8 and the expression from #User9. As of May 11, 2023, the news in Fig. 1 has 10.2K retweets

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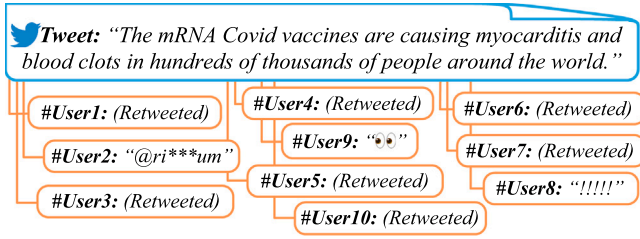


Fig. 1. An illustrative example of a piece of news spreading by retweeting.

without comments, but only 380 retweets with comments. Therefore, the challenge in real-world fake news detection is how to exploit non-textual auxiliary information from social context for accurate results, which we refer to as *challenge #1*.

To address the *challenge #1*, some models utilizing non-textual structural information have been proposed. For example, some models construct propagation graphs for tweets and retweets (Bian et al., 2020), while others employ heterogeneous graphs for news and users (Huang, Yu, Wu, & Wang, 2020). These approaches have demonstrated notable performance in identifying fake news, as the extracted structural information aligns well with the network formed by news spreading or news-user interaction in social media. However, these methods still have a limitation: they typically capture pairwise relations in social networks. In real-world social networks, there are numerous high-order and intricate relations that these methods fail to capture beyond pairwise relations. For instance, a pairwise relation can represent the relationship between a user and a single piece of news he or she posts, but it cannot represent the relations among multiple news posts by the same user. Thus, extracting high-order relations from social networks to gather in-depth information for enhancing fake news detection presents *challenge #2*.

In this work, to address the *challenge #1*, we leverage user information to complement the plain text of news content. Given the participatory and user-driven nature of news dissemination on online social networks (Vavilis, Petković, & Zannone, 2014), we argue that users' engagement and dissemination patterns may vary between fake and real news, providing valuable insights for fake news detection. Specifically, to incorporate non-textual information from the social context, we aim to extract the inherent attributes of users as well as the structures of both news and users during news spreading. To tackle the *challenge #2*, we employ the concept of hypergraph to capture intricate high-order relations between news and users (Bretto, 2013). Hypergraphs are generalizations of simple graphs that can connect an arbitrary number of nodes in an edge. As depicted in Fig. 2, a simple graph only connects two nodes in an edge and its adjacency matrix illustrates the pairwise relation between nodes. However, the incidence matrix of a hypergraph demonstrates that multiple nodes are assigned to one edge, enabling the extraction of high-order information. Therefore, in this work users are abstracted as nodes, and a hyperedge is formed when multiple users engage with the same piece of news through retweeting. This hypergraph construction allows us to capture the intricate relation between a piece of source news and the users who participated in its dissemination within each hyperedge, enabling the learning of distinctive information for fake news detection. If a user participates in the same news multiple times, this user is still abstracted as one node. Within this hypergraph, node attributes represent user properties related to credibility, while hyperedge attributes encompass news textual content.

To comprehensively learn the latent high-order correlation between news and users in our constructed hypergraph and overcome the two challenges, we propose a method based on **Hypergraph** neural networks for **Detecting Fake** news, abbreviated as Hy-DeFake. Hy-DeFake consists of four main parts, utilizing the attributed hypergraph as

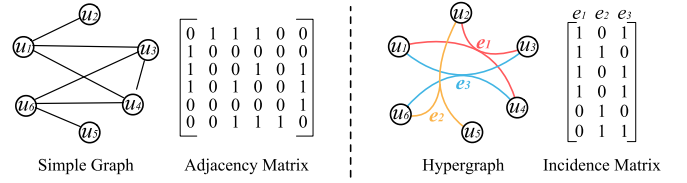


Fig. 2. Difference between simple graph and hypergraph.

input: (1) News semantic channel: it updates hyperedge features by learning semantic embeddings of news contents, as each hyperedge represents a piece of news; (2) User credibility channel: it learns node features through embedding both user credibility information and the high-order structural information between news and users; (3) Mutual information-based feature fusion: it incorporates the semantic embeddings of news (i.e., hyperedges) and the credibility embeddings of relevant users (i.e., nodes) based on maximizing their mutual information; (4) Fake news detection: it models our task as hyperedge classification and classifies the news based on the integrated embeddings. Specifically, Hy-DeFake simultaneously trains the news semantic channel and the user credibility channel to acquire diverse information for identifying fake news. The news semantic channel leverages a language model to exploit semantic embeddings of news. The user credibility channel employs a hypergraph autoencoder, where the encoder is a hypergraph convolution network, that learns the high-order information between nodes and hyperedges. Through this process, the two channels learn the semantic embeddings of news and user embeddings, which capture both user credibility and high-order correlation between news and users, respectively. This work presents the following key contributions:

- We introduce an approach by constructing an attributed hypergraph to represent the process of news spreading in online social networks. By abstracting fake news detection as hyperedge classification, we capture the intricate high-order relation between news and users in social contexts, enabling us to achieve accurate results.
- The proposed Hy-DeFake utilizes hypergraph neural networks for fake news detection. It effectively captures the credibility information of users and the high-order correlation between news and users. Both of these aspects provide distinctive information that contributes to fake news detection.
- Extensive experiments demonstrate that Hy-DeFake generally surpasses nine baseline methods on four real-world datasets from different domains.
- We verify the positive correlation between news authenticity and user credibility. Users who spread fake news exhibit more intensive interaction compared to those who spread real news, resulting in the formation of a denser community.

This paper consists of the following parts. Section 2 reviews the relevant literature on hypergraph learning, fake news detection, and the distinctions between Hy-DeFake and existing methods. The relevant definitions and hypergraph construction are formulated in Section 3. Following that, Section 4 presents each part of the proposed framework, and Section 5 shows experimental evaluations. Finally, we draw conclusions in this work in Section 6.

2. Related work

To give an overview of the relevant research, this section first reviews the work on hypergraph learning and its applications, then further discusses the work on fake news detection.

2.1. Hypergraph learning

The development of graph neural networks (GNN) (Wu et al., 2021) has demonstrated its success in various tasks and has garnered significant attention for its ability to uncover structural relations (Su et al., 2022; Wu, Sun, Zhang, Xie & Cui, 2022). However, traditional graph learning methods primarily focus on pairwise connections, limiting their capacity to express complex high-order relations that extend beyond pairwise associations. Therefore, as a promising and flexible technique for modeling complex high-order relations, hypergraph learning is emerging recently. The first work that applies GNNs to hypergraphs is HGNN (Feng, You, Zhang, Ji, & Gao, 2019). To leverage complex and high-order relations for better representation learning, it designs a hyperedge convolution operation. HyperGCN (Yadati et al., 2019) trains a GCN on hypergraph by adopting tools from the spectral theory of hypergraphs and extending its faster variant. Hyper-SAGNN (Zhang, Zou, & Ma, 2019) develops a self-attention based GNN for homogeneous and heterogeneous hypergraphs. DHGNN (Dynamic Hypergraph Neural Networks) (Jiang, Wei, Feng, Cao, & Gao, 2019) is a framework that exploits adjusted feature embeddings to dynamically update hypergraph structure. HyperGCL (Wei et al., 2022) is a hypergraph generative model that applies contrastive learning for robust and fair hypergraph representation learning.

Due to its capacity to capture complicated high-order relations, hypergraph neural networks have found applications in various domains. For instance, SHINE (Luo, 2022) is a subhypergraph inductive neural network for simultaneously capturing functional relations for genes and pathways. HHGR (Zhang, Gao, et al., 2021) and DH-GCN (Han, Tao, Tang, & Xia, 2022) are two examples of exploiting hypergraph neural networks for social recommendation. GroupNet (Xu, Li, Ni, Zhang, & Chen, 2022) introduces a multiscale hypergraph neural network for trajectory prediction. We can conclude that a hypergraph neural network is suitable for numerous real-world scenarios, particularly in complex social networks. However, its application in the domain of detecting fake news in online social networks is relatively limited. Available approaches utilizing hypergraphs for fake news detection are quite limited (Ding, Wang, Li, Li, & Liu, 2020; Jeong et al., 2022). HyperGAT (Ding et al., 2020) primarily focuses on news content, lacking the integration of high-order relations of both users and news. HGFND (Jeong et al., 2022) requires numerous types of hyperedges, increasing the difficulty of dataset construction and impacting real-world practicality. Moreover, it still relies on vanilla GNNs to learn hypergraph representations and is limited in capturing latent high-order relations. Thus, this work is proposed to address such issues.

2.2. Fake news detection

2.2.1. Text-based detection

From the perspective of news contents, text-based approaches detect fake news by distinguishing linguistic features and writing styles between real and fake news. Pérez-Rosas et al., for instance, Pérez-Rosas et al. (2018) introduced a text-based method focusing on learning linguistic features to identify fake news. In FakeBERT (Kaliyar, Goswami, & Narang, 2021), Kaliyar et al. integrated BERT (Devlin, Chang, Lee, & Toutanova, 2019) with multiple parallel blocks of 1d-CNN for fake news detection. Apart from analyzing the textual contents of the news itself, auxiliary textual information is also integrated to enhance fake news detection. deFEND (Shu et al., 2019), for example, adopts a hierarchical co-attention mechanism to capture the explainable sentences from both news and the relevant comments. Similarly, STANKER (Rao et al., 2021) utilizes auxiliary features through comments and leverages the level-grained attention-masked BERT model to uncover fake news. Furthermore, mining stance or emotion in news content can provide side information as well. For example, Zhang et al. (Zhang, Cao, et al., 2021) specifically examined the impact of emotional cues in real and fake news, and incorporated dual emotion features to represent both

individual emotions and the relationship between them. Yang et al. proposed two tree-structured models (TD-MIL and BU-MIL) (Yang, Ma, Lin & Gao, 2022) to simultaneously verify rumor claims and detect the stances behind the relevant posts. Sheng et al. (Sheng et al., 2022) consider both macro and micro news environments for fake news identification. FTT (Hu et al., 2023) detects fake news by discovering topics in the news and forecasting temporal trends.

For considering users' engagement and opinion, CSI (Ruchansky, Seo, & Liu, 2017) is an RNN-based model integrating three characteristics of news and users, *i.e.*, news text, users' response, and the source of news. While text-based models utilize textual information and auxiliary features for fake news detection, they commonly analyze each news piece in isolation, overlooking the intrinsic news-spreading mechanism. As a result, they may neglect valuable propagation information. These methods often lack user information which is crucial for understanding the social context to identify fake news accurately. Consequently, the performance of these approaches becomes limited.

2.2.2. Graph-based detection

Due to the compatibility between the propagation nature of news and the structural property of graphs, graph learning shows high-quality results in fake news detection. In graph-based methods, researchers typically extract structural relations of words or sentences, construct news propagation graphs or incorporate social contexts (such as user information) to detect fake news from both textual and structural information perspectives. For instance, Yao et al. used GCN (Kipf & Welling, 2017) to learn news-level structure information. They constructed graphs to represent individual news pieces (Yao, Mao, & Luo, 2019). Wu, Liu, Xu and Wu (2022) developed a causal graph to depict the relations between news and evidence in a counterfactual inference framework. TraceMiner (Wu & Liu, 2018) is an LSTM-RNN-based model to classify propagation traces of news. Additionally, Ma et al. (2018) created a tree-structured graph based on tweets and utilized recursive neural networks to represent the graph in both top-down and bottom-up fashion. Similarly, Bi-GCN (Bian et al., 2020) is a bi-directional graph neural network that understands patterns of news propagation and dispersion by GCN (Kipf & Welling, 2017). To sum up, graph-based approaches have the ability to represent the textural and structural properties of debunking fake news.

To fully harness the potential of graph structure for accurate results, researchers also incorporate structural side information. CompareNet (Hu et al., 2021) exploits knowledge graphs to detect fake news. For capturing various connections, Huang et al. (2020) and HGAT (Ren & Zhang, 2021) are two works constructing heterogeneous information networks (HINs). Sethi (2017) verified alternative facts and constructed an argumentation graph for crowdsourcing the fake news verification. DistrustRank (Woloszyn & Nejdil, 2018) obtains a trust rank of news websites by constructing a graph and calculating the similarity between the untrustworthy sites and the existence of controversial subjects.

For mining user information such as user credibility or reputation in social contexts, UPFD (Dou, Shu, Xia, Yu, & Sun, 2021) considers user preferences by summarizing historical posts, Us-DeFake (Su, Yang, Wu, & Zhang, 2023) captures news-user relations in a dual-layer graph, PESTO (Min & Ananiadou, 2023) defines a post-user fusion network to model the rumor patterns from both post diffusion and user social networks, and SGCP (Che, Pan, Leung, Cao, & Yan, 2023) is a sparse and graph-regularized decomposition in which a third-order tensor is defined to preserve the complex relations among news and users. DETECTIVE (Tschitschek, Singla, Gomez Rodriguez, Merchant, & Krause, 2018) utilizes Bayesian inference and accurates users' flagging to detect fake news based on crowd signals. HCS (Freire, da Silva, & Goldschmidt, 2021) combines crowd signals and meta-learning to avoid the dependence on the explicit opinion of users for fake news detection. MFAN (Zheng et al., 2022) integrates texts, visions, and social graph

features in a unified framework, to obtain complementary relations between different modalities. Furthermore, Mehta et al. (Mehta, Pacheco, & Goldwasser, 2022) used inference operators to reveal interactions between news content and user engagement patterns. SureFact (Yang, Wang, et al., 2022) is based on multiple heterogeneous subgraphs that extract diverse information from claims, posts, keywords, and users. Similarly, Jin et al. constructs a claim-evidence graph by news articles, posts, and users, and then proposes (Jin et al., 2022) a fine-grained reasoning model FinerFact. PSIN (Min et al., 2022) models the post and user interactions through a divide-and-conquer strategy to detect fake news. DECOR (Wu & Hooi, 2023) refines social graphs by degree-corrected stochastic blockmodels for accurate fake news detection. Albeit the variety of methods considers user information in social contexts, they are inadequate in modeling the complex high-order relations between news and users. Therefore, we propose Hy-DeFake to address the aforementioned problems.

2.2.3. Early detection

Detecting and mitigating the spread of fake news before it causes harm is crucial. Therefore, SMAN (Yuan, Ma, Zhou, Han, & Hu, 2020) detects fake news in the early dissemination stages by capturing the credibility of news, publishers, and users based on a structure-aware multi-head attention network. Propagation2Vec (Silva, Han, Luo, Karunasekera, & Leckie, 2021) is a method based on propagation networks. It utilizes a hierarchical attention mechanism to assign varying levels of attention to nodes and cascades within propagation patterns. This approach enables the reconstruction of knowledge, facilitating explainable early detection. PPC RNN+CNN (Liu & Wu, 2018) detects fake news at an early stage by classifying news propagation paths, which are modeled as multivariate time series and a time series classifier combining recurrent and convolutional networks are designed to identify fake news. Based on time-aware crowd signals, TCS (Cavalcante, Freire, Goldschmidt, & Justel, 2024) explores the temporal nature of news propagation for fake news early detection. It incorporates users' reputations derived from their past public behavior in news dissemination. PTK (Ma, Gao, & Wong, 2017) (Propagation Tree Kernel) is a kernel-based method for early detection, designed to distinguish different types of rumors by analyzing high-order patterns through similarities in their propagation tree structures.

3. Preliminaries

In this section, we begin by describing the notions and definitions utilized in this paper, and then present how a social network with news and users is modeled as a hypergraph. Finally, we state our problem of detecting fake news.

3.1. Notations and definition

Let $\mathcal{N} = \{n_1, n_2, \dots, n_t\}$ denote the news set, where t represents the number of news and n_i represents the textual contents of news i . Let $\mathcal{U} = \{u_1, u_2, \dots, u_m\}$, where m is the number of users. The corresponding user attributes are defined as $\mathbf{X} \subseteq \mathbb{R}^{m \times d}$, where d refers to the dimension of user attribute vectors.

Hypergraph Definition: Hypergraph is the generalized graph, it can be formulated as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ and $\mathbf{X}$ indicate the node set and the corresponding node attributes. $\mathcal{E}$ signifies the set of hyperedges, where each hyperedge connects to a subset of nodes rather than two nodes. The diagonal matrices of edge degrees and node degrees can be represented as $\mathbf{D}_e$, $\mathbf{D}_v$, respectively. $\mathbf{H} = |\mathcal{V}| \times |\mathcal{E}|$ denotes incidence matrix, where $h(v, e) = 1$ if the node v is on hyperedge e (i.e., $v \in e$), else $h(v, e) = 0$.

3.2. Hypergraph construction

To model the relation between each news piece and the users involved in its spreading in online social networks for debunking fake

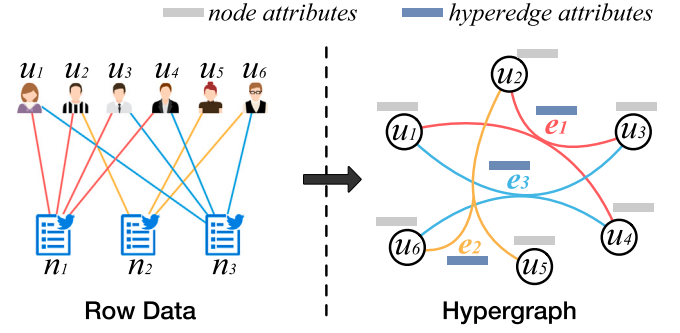


Fig. 3. Hypergraph construction.

news, we define an attributed hypergraph as $\mathcal{G} = (\mathcal{U}, \mathcal{E}, \mathbf{X}, \mathcal{N})$ to represent the beyond pairwise relations. Here the node set $\mathcal{U}$ denotes the users who are involved in news spreading. The hyperedge set $\mathcal{E}$ denotes the interactions between news and users, where each hyperedge e_i represents a piece of news i and can connect more than two users involved in spread this news. $\mathbf{X}$ and $\mathcal{N}$ represents attributes of nodes (i.e., users) and hyperedges (i.e., news), respectively. The node attributes of users are credibility-related properties, while the hyperedge attributes are news textual contents. Fig. 3 shows the changes in row data after hypergraph construction.

Thus, the task of this work is abstracted as distinguishing news in a hypergraph, i.e., hyperedge classification task. Treating fake news detection as a binary classification task, we associate a binary label $y \in \{0, 1\}$ to each hyperedge e_i in a hypergraph. A label of 0 is assigned to indicate that the corresponding hyperedge represents real news, whereas a label of 1 is assigned to indicate fake news. Thus, the hypergraph for fake news detection can be denoted as $\mathcal{G} = (\mathcal{U}, \mathcal{E}, \mathbf{X}, \mathcal{N}, \mathcal{Y})$, where $\mathcal{Y}$ represents the label set assigned to the news.

4. Methodology

This section provides a detailed explanation of Hy-DeFake. The model takes our constructed hypergraph as input. To fully capture credibility information from users and semantic information from news, we design two channels to learn user and news information individually. As depicted in Fig. 4, Hy-DeFake comprises four parts: (1) the news semantic channel, (2) the user credibility channel, (3) the mutual information-based feature fusion, and (4) the fake news detection. The news semantic channel adopts a language model to update hyperedge features and comprehend news semantics. The user credibility channel exploits an unsupervised hypergraph autoencoder to acquire credibility features of users, preserving the high-order information between news and users. Then, the mutual information-based feature fusion module aggregates node features of users involved in each piece of news to obtain representative credibility embeddings of users. This module also maximizes the mutual information of news embeddings and user embeddings in high-dimensional space to capture positive relations between news and users, optimizing the final hyperedge embeddings. Finally, the fake news classifier distinguishes the news based on integrated embeddings.

4.1. News semantic channel: Updating hyperedge features

When detecting fake news, news contents play a crucial role in discriminating lexical features or writing styles of fake news from real ones. Thus, we devise a news semantic channel and take the source news contents as the hyperedge attributes in the input hypergraph for our proposed model, where each hyperedge represents a piece of news. To update the textual features of news to contain semantic information,

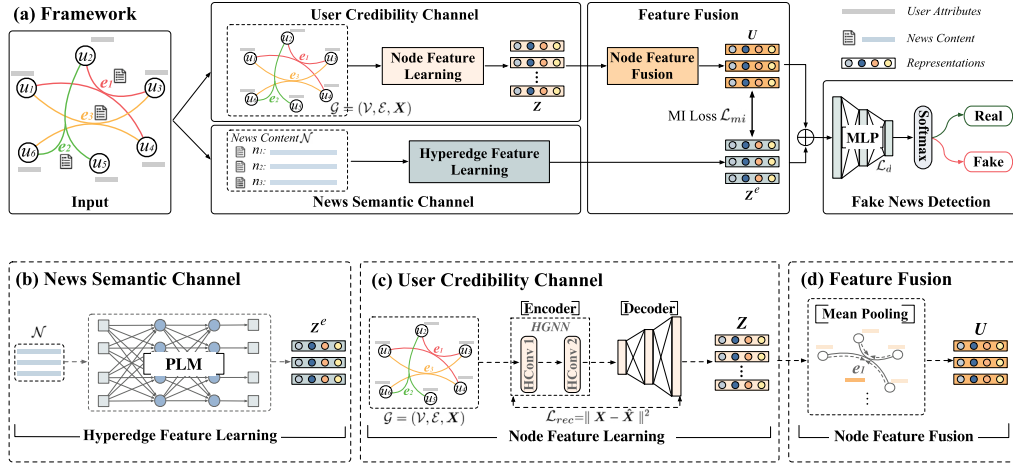


Fig. 4. The Framework of Hy-DeFake. (a) **Framework** shows the overall architecture of Hy-DeFake. It takes a hypergraph as input and classifies news simultaneously. Firstly, Hy-DeFake designs two channels for learning features from news and users, respectively. Hyperedge attributes (i.e., news content) are fed into the news semantic channel to learn the semantic features of news. Hypergraph with node attributes is fed into the user credibility channel to learn node features with credibility information of users and high-order structural information between news and users. After that, a mutual information-based feature fusion module fuses hyperedge features and representative node features to obtain integrated embeddings. Finally, the fused embeddings are fed into the fake news detection component to classify the news into real and fake. (b) **News Semantic Channel** shows the learning process of hyperedge attributes $\mathcal{N}$ to update news features Z^e in a pre-trained language model (PLM). (c) **User Credibility Channel** adopts a hypergraph neural network (HGNN) in Autoencoder architecture to learn the node features Z , where ‘HConv’ represents hypergraph convolution. (d) **Feature Fusion** fuses Z on each hyperedge to aggregate the representative user embeddings U for further fusion with news embeddings.

we utilize a pre-trained language model $\mathcal{M}$ to derive hyperedge features. Here we employ RoBERTa (Liu et al., 2019) which is a robustly optimized BERT pre-trained model. The process is as follows:

$$z_i^e = \mathcal{M}(n_i, \forall n_i \in \mathcal{N}), \quad (1)$$

where z_i^e stands for the updated hyperedge feature of a piece of news n_i . Through the language model’s fine-tuning, the textual contents of news can be processed as vector embeddings that embed semantic information of news. In Hy-DeFake, since each piece of news is abstracted as a hyperedge, we take the textual embeddings as the hyperedge features Z^e , to provide semantic information in hypergraph for accurate fake news detection results.

4.2. User credibility channel: Learning node features

Users are indispensable creators and disseminators of news in on-line social media, influencing the spread of information through their actions. They may exhibit differing behaviors toward trustworthy and untrustworthy news. Credible users tend to propagate reliable news, often showing restraint or skepticism toward uncertain or false information. Conversely, certain users may be more likely to engage with or spread misleading news. In this work, Hy-Fake does not explicitly classify users as malicious or benign. Instead, it captures patterns related to user credibility attributes in an unsupervised manner to assist in fake news identification. Here the user credibility channel is designed to explore the correlation between users and the news they disseminate, learning not only user credibility features but also high-order, complex relationships between news and users in real-world social networks.

For learning the complicated correlation between news and users with user credibility information, we propose a hypergraph Autoencoder consisting of an encoder and a decoder. Due to the limitation of traditional graph encoders in encoding complex correlations more than pairwise connections, Hy-DeFake adopts a hypergraph convolution network (Feng et al., 2019) as the encoder. It designs a hyperedge convolution operation for learning the high-order latent correlation between nodes and hyperedges. Specifically, it performs node-edge-node transform, which can capture user-news-user relations and refine the node features of user credibility using the hypergraph structure. The hypergraph convolution is defined by

$$Z^{(l)} = \sigma(\mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{W}_e^{-1} \mathbf{H}^T \mathbf{D}_v^{-1/2} Z^{(l-1)} \Theta^{(l-1)}), \quad (2)$$

where $\mathbf{W}$ signifies the learnable parameter during the training process, $Z^{(0)} = \mathbf{X}$, Θ is the filter, and σ indicates the nonlinear activation function.

Through the aforementioned encoding process, the encoder maps the input data to the embeddings of users which contain user credibility information and high-order correlation between users and news. For unsupervised training of this channel, the decoder maps the embeddings back to reconstruct the input user attributes. The formulation is as follows:

$$\begin{aligned} \hat{x}_i^{(1)} &= \sigma(\tilde{\mathbf{W}}^{(1)} z_i + b^{(1)}), \\ &\dots \\ \hat{x}_i^{(k)} &= \sigma(\tilde{\mathbf{W}}^{(k)} \hat{x}_i^{(k-1)} + b^{(k)}). \end{aligned} \quad (3)$$

In Eq. (3), z_i represents the i th node’s latent representation learned by Eq. (2), $\hat{x}_i^{(k)}$ is the desired reconstructed attribute of node i , $\{\tilde{\mathbf{W}}^{(1)}, \dots, \tilde{\mathbf{W}}^{(k)}, b^{(1)}, \dots, b^{(k)}\}$ are the parameters of the decoder with k layers. The training of hypergraph Autoencoder is to minimize the reconstruction error between the input user attributes and the reconstructed user features. The loss function is as follows:

$$\begin{aligned} \mathcal{L}_{rec} &= \sum_{i=1}^m \|x_i - \hat{x}_i\|^2 = \|\mathbf{X} - \hat{\mathbf{X}}\|^2 \\ &= \|\mathbf{X} - \mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{W}_e^{-1} \mathbf{H}^T \mathbf{D}_v^{-1/2} \mathbf{X}\|^2. \end{aligned} \quad (4)$$

4.3. Mutual information-based feature fusion

Through the learning process in the two channels outlined above, Hy-DeFake extracts semantic features of news embedded in hyperedges, along with user credibility features and higher-order relations represented by nodes in the hypergraph. Previous work (Li, Zhao, Yang, Sheng, & Wu, 2020; Su et al., 2023) has demonstrated a positive correlation between news authenticity and user credibility, which can be utilized in Hy-DeFake to learn distinctive representations by aligning associated news and users in the training process while separating unassociated pairs. To leverage this relationship, we employ a mutual information (MI)-based feature fusion module to integrate these semantic features, credibility features, and higher-order relationships, yielding distinctive embeddings for the final task. First, we fuse the features of users associated with each news article to derive a representative user credibility embedding for that news. Next, we combine

the news semantic embeddings with the representative user credibility embeddings using an MI loss, resulting in integrated embeddings specifically designed for fake news detection.

Each hyperedge has multiple nodes, indicating that the news is propagated by multiple users. We define $\mathcal{U}_i^s \subseteq \mathcal{U}$ to represent a subset of users involved in news i . To obtain the representative embedding of users $\mathcal{U}_i^s$, we aggregate credibility embeddings of users involved in news i by the element-wise mean pooling, which can be expressed as follows:

$$\mathbf{u}_i = \text{MEAN}\{\mathbf{z}_j, \forall j \in \mathcal{U}_i^s \subseteq \mathcal{U}\}, \quad (5)$$

where $\mathbf{u}_i$ is the representative credibility embedding of users involved in the dissemination of news i , and $\mathbf{z}_j$ is the credibility embedding of j th user in $\mathcal{U}_i^s$. Thus, we obtain the mutual representative user credibility matrix $\mathbf{U}$.

Subsequently, to capture the latent correlation between news and users, we introduce a mutual information loss to identify the latent dependencies between news features and their corresponding representative user features. The loss function is maximized by the following equation:

$$\mathcal{L}_{mi} = \frac{1}{t} \sum_{i=1}^t \left[\mathcal{I}(\mathbf{z}_i^e; \mathbf{u}_i) + \mathcal{I}(\mathbf{u}_i; \mathbf{z}_i^e) \right], \quad (6)$$

where t indicates the number of news, $\mathbf{z}_i^e$ and $\mathbf{u}_i$ are the embedding of news i and the representative embedding of users participated in spread of news i , respectively. The mutual information can be calculated by:

$$\mathcal{I}(\mathbf{z}_i^e; \mathbf{u}_i) = \mathbb{E}_{(\mathbf{z}_i^e, \mathbf{u}_i) \sim p(\mathbf{z}_i^e, \mathbf{u}_i)} \left[\log \frac{p(\mathbf{z}_i^e, \mathbf{u}_i)}{p(\mathbf{z}_i^e)p(\mathbf{u}_i)} \right]. \quad (7)$$

Due to the assumption of positive correlation, our method maximizes the loss function in Eq. (6). This maximization allows Hy-DeFake to enhance mutual information in two channels, facilitating the sharing of positive relations between credible users and real news, as well as uncredible users and fake news, respectively. Consequently, Hy-DeFake effectively learns distinctive embeddings for the task of identifying fake news.

4.4. Fake news detection

After obtaining the final embeddings that fuse news semantic information and user credibility information, such integrated embeddings can be regarded as hyperedge embeddings, because each hyperedge represents an individual news piece. The task of discovering fake news becomes a hyperedge classification. In Hy-DeFake, the integrated hyperedge embeddings act as input data for the hyperedge classifier. We pass the hyperedge embeddings through a multilayer perceptron (MLP) followed by a softmax layer for final news prediction, which is formulated as follows:

$$\mathbf{z}_i^o = f(W^{(k)}(\mathbf{z}_i^e \oplus \mathbf{u}_i) + b^{(k)}), k = 1, \dots, K \quad (8)$$

$$\hat{y} = \text{SOFTMAX}(\mathbf{z}_i^o), \quad (9)$$

where $\mathbf{z}_i^o$ is the output embedding of news i after MLP, $\oplus$ denotes the concatenation operation, $W^{(k)}$ and $b^{(k)}$ denote the parameters on layer k . $\hat{y}$ represents the predicted label. For each piece of news, i.e., each hyperedge, this component's objective is to minimize the cross-entropy loss:

$$\mathcal{L}_d = -y \log \hat{y} - (1 - y) \log (1 - \hat{y}), \quad (10)$$

where $y \in \{0, 1\}$ represents the ground truth label of news. $\hat{y}$ denotes the predicted value, which indicates the probability that the news is fake. In Hy-DeFake, we combine the loss functions of user credibility encoding, MI-based fusion of news and users, and hyperedge classification in a unified framework. The total loss function to be optimized becomes:

$$\mathcal{L} = \mathcal{L}_{rec} + \mathcal{L}_d + \alpha \mathcal{L}_{mi}, \quad (11)$$

where α is a coefficient to control the balance of objectives. The pseudo-code for the training of Hy-DeFake is illustrated in Algorithm 1.

Algorithm 1 The training process of Hy-DeFake

Input: $\mathcal{N}$: news content; $\mathcal{G} = (\mathcal{U}, \mathcal{E}, \mathbf{X})$: hypergraph of news and users; $\mathcal{Y}_{tr}$: training labels of news on hyperedges.

Parameters: α : coefficient balancing the loss function.

Output: $\mathbf{Z}^o$: the learned news representations on hyperedges.

```

1: while epoch do
2:   Update  $\mathbf{Z}^e$  of hyperedges by Eq. (1).
3:   Update  $\mathbf{Z}$  of nodes by Eq. (2).
4:   Reconstruction of user attributes  $\hat{\mathbf{X}}$  by Eq. (3).
5:   Obtain the representative user embeddings  $\mathbf{U}$  by Eq. (5).
6:   Fusion of  $\mathbf{Z}^e$  and  $\mathbf{U}$  to obtain  $\mathbf{Z}^o$ .
7:   Compute the reconstruction loss  $\mathcal{L}_{rec}$  by Eq. (4).
8:   Compute the MI loss  $\mathcal{L}_{mi}$  by Eq. (6).
9:   Compute the cross-entropy loss  $\mathcal{L}_d$  by Eq. (10).
10:  Backward propagation via total loss  $\mathcal{L}$  by Eq. (11).
11: end while
12: return  $\mathbf{Z}^o$  for classification.

```

Table 1

The dataset statistics.

Datasets	Nodes (Users)	Hyperedges (News)	Labels
PolitiFact	27,682	635	R: 344/F: 291
ReCOVery	36,741	2,026	R: 1,364/F: 662
MM-COVID	18,672	3,992	R: 1,894/F: 2,098
Gossipcop	72,083	10,084	R: 5,444/F: 4,640

5. Experiments

This section validates the performance of Hy-DeFake by presenting experimental conduction and results. Specifically, after introducing the experimental setup, we assess the overall performance of Hy-DeFake by comparing it against nine popular baseline algorithms, and analyze the effectiveness of each module in Hy-DeFake, respectively. Further, we verify the explainability, robustness, and efficiency of Hy-DeFake.

5.1. Experimental setup

5.1.1. Datasets

To investigate the efficacy of Hy-DeFake on fake news detection, we choose four real-world datasets, named PolitiFact [Shu, Mahudeswaran, Wang, Lee, and Liu \(2020\)](#), ReCOVery [\(Zhou, Mulay, Ferrara, & Zafarani, 2020\)](#), MM-COVID [\(Li, Jiang, Shu, & Liu, 2020\)](#), and Gossipcop [Shu et al. \(2020\)](#). These datasets cover various domains, with PolitiFact focusing on political news, ReCOVery and MM-COVID containing data related to COVID-19, and Gossipcop collecting news related to entertainment. All the datasets include fact-checked source news, and the relevant social context, i.e., related tweets along with users who participate in the news spread. To identify fake news by incorporating the natural pattern of news outbreak and spreading, we used Twitter API¹ to crawl user information, which includes “followers_count”, “friends_count”, “listed_count”, “verified”, “statuses_count”, and “favourites_count”. Table 1 summarizes the dataset statistics, where each “Hyperedge” represents individual news, and “Nodes” represent the users participating in the news spreading. The labels assigned to the hyperedges are “real” and “fake”.

5.1.2. Baselines and metrics

In our evaluation, we employ four frequently employed metrics, i.e., Accuracy (ACC), Precision (Pre), Recall (Rec), and F1 score (F1), to compare the performance of our method with nine popular baseline algorithms. Here is a brief introduction to the baseline algorithms:

¹ <https://developer.twitter.com/en/docs/twitter-api>

- TextCNN (Kim, 2014) is a CNN (convolutional neural network) based method to classify sentences, it builds CNNs and utilizes convolution filters on *word2vec* of news contents to capture textual features in different granularity.
- HAN (Yang et al., 2016) (hierarchical attention network) is proposed to classify documental content. It adopts an attention mechanism at both the sentence and word levels to encode news content.
- BERT (Devlin et al., 2019) is a prominent pre-trained language model for natural language processing (NLP) tasks, that utilizes a transformer-based architecture with a bidirectional encoder and self-attention heads. Here we fine-tune the BERT model specifically for identifying fake news.
- TextGCN (Yao et al., 2019) is a graph-based approach that extends GCN (Kipf & Welling, 2017) to perform text classification. It captures semantic relationships between words and sentences in the graph representation.
- HyperGAT (Ding et al., 2020) is a hypergraph-based inductive method for textual data. It constructs a document-level hypergraph attention network for each piece of news to learn the textual representations in fake news detection.
- DualEmo (Zhang, Cao, et al., 2021) is the fake news detection model that mines dual emotions from publishers and users in a social context. It identifies the emotional gap between source news and relevant comments to distinguish fake news.
- HGFND (Jeong et al., 2022) is a hypergraph learning-based method to capture group-wise interactions among news. It leverages a dual-level attention mechanism to represent important news relations.
- Llama 2 (Touvron et al., 2023) is a collection of fine-tuned and pre-trained large language models (LLMs). Here we utilize the chat model to learn the representations of news and then identify real and fake news by classifier.
- SMAN (Yuan et al., 2020) detects fake news by capturing the credibility of news, publishers, and users based on a structure-aware multi-head attention network.

5.1.3. Experimental setup and implementation

The datasets are split into train-validation-test sets with a ratio of 70%-10%-20%. In Hy-DeFake, the embedding size is set to 768. We use two-layer HGNN (Feng et al., 2019) as encoder and a full-connected decoder to reconstruct user attributes with the dimension of 6. We use a three-layer MLP as the hyperedge classifier with hidden layer dimensions of 512 and 128. The model optimization is performed using the Adam optimizer with a learning rate of 0.001 for all datasets. The value of α is set to -0.01 . The number of epochs is set to 600. The evaluation is conducted using 5-fold cross-validation.

5.2. Performance

The performance of Hy-DeFake, along with nine popular baseline methods, is evaluated on four real-world datasets. The results are presented in Tables 2, 3, 4 and 5 with the optimal results highlighted in gray. As shown in these tables, the performance of Hy-DeFake is superior to that of the baseline methods on three datasets, and achieves sub-optimal or optimal performance on one dataset, *i.e.*, Gossipcop. This illustrates that the embeddings of Hy-DeFake that incorporate news semantics and user credibility learn sufficient information compared to other methods. Further, capturing high-order correlation between news and users is effective for identifying fake news.

Regarding the baseline performances, Llama 2 exhibits sub-optimal performance, as it is a large language model with billions of parameters. The version of Llama 2 utilized in our experiments is the chatbot model, which conducts text generation inference and demonstrates a heightened understanding of semantics in news content. The performance of HGFND is also satisfactory, as it considers high-order relations in

Table 2

Overall performance for fake news detection on PolitiFact.

Methods	ACC	Pre	Rec	F1
TextCNN	0.506 ± 0.038	0.302 ± 0.067	0.529 ± 0.115	0.378 ± 0.067
HAN	0.521 ± 0.015	0.416 ± 0.052	0.529 ± 0.055	0.465 ± 0.049
BERT	0.786 ± 0.031	0.869 ± 0.034	0.808 ± 0.046	0.841 ± 0.025
TextGCN	0.740 ± 0.032	0.741 ± 0.034	0.741 ± 0.033	0.739 ± 0.032
HyperGAT	0.869 ± 0.021	0.871 ± 0.022	0.869 ± 0.022	0.868 ± 0.022
DualEmo	0.830 ± 0.027	0.831 ± 0.027	0.825 ± 0.028	0.824 ± 0.028
HGFND	0.884 ± 0.028	0.924 ± 0.002	0.891 ± 0.049	0.877 ± 0.028
Llama 2	0.913 ± 0.021	0.919 ± 0.016	0.907 ± 0.025	0.910 ± 0.021
SMAN	0.895 ± 0.009	0.897 ± 0.009	0.899 ± 0.017	0.892 ± 0.009
Hy-DeFake	0.929 ± 0.009	0.939 ± 0.009	0.925 ± 0.013	0.928 ± 0.011

Table 3

Overall performance for fake news detection on ReCOVery.

Methods	ACC	Pre	Rec	F1
TextCNN	0.406 ± 0.041	0.266 ± 0.060	0.691 ± 0.016	0.381 ± 0.066
HAN	0.385 ± 0.028	0.224 ± 0.030	0.675 ± 0.065	0.336 ± 0.041
BERT	0.756 ± 0.026	0.778 ± 0.028	0.749 ± 0.035	0.751 ± 0.013
TextGCN	0.701 ± 0.024	0.688 ± 0.078	0.616 ± 0.059	0.597 ± 0.094
HyperGAT	0.648 ± 0.009	0.581 ± 0.014	0.580 ± 0.014	0.579 ± 0.014
DualEmo	0.818 ± 0.031	0.781 ± 0.037	0.763 ± 0.041	0.769 ± 0.037
HGFND	0.923 ± 0.011	0.919 ± 0.013	0.908 ± 0.009	0.917 ± 0.005
Llama 2	0.938 ± 0.007	0.930 ± 0.010	0.922 ± 0.009	0.926 ± 0.009
SMAN	0.919 ± 0.012	0.912 ± 0.023	0.908 ± 0.011	0.909 ± 0.015
Hy-DeFake	0.952 ± 0.009	0.958 ± 0.010	0.938 ± 0.019	0.946 ± 0.014

Table 4

Overall performance for fake news detection on MM-COVID.

Methods	ACC	Pre	Rec	F1
TextCNN	0.496 ± 0.016	0.535 ± 0.015	0.480 ± 0.016	0.506 ± 0.010
HAN	0.502 ± 0.009	0.419 ± 0.053	0.480 ± 0.019	0.446 ± 0.036
BERT	0.879 ± 0.019	0.873 ± 0.020	0.881 ± 0.022	0.877 ± 0.018
TextGCN	0.787 ± 0.157	0.716 ± 0.258	0.744 ± 0.198	0.696 ± 0.259
HyperGAT	—	—	—	—
DualEmo	0.883 ± 0.021	0.881 ± 0.024	0.884 ± 0.021	0.881 ± 0.022
HGFND	0.905 ± 0.014	0.915 ± 0.010	0.909 ± 0.013	0.904 ± 0.014
Llama 2	0.947 ± 0.008	0.945 ± 0.009	0.935 ± 0.012	0.940 ± 0.011
SMAN	0.898 ± 0.011	0.910 ± 0.009	0.904 ± 0.013	0.898 ± 0.011
Hy-DeFake	0.956 ± 0.004	0.956 ± 0.004	0.956 ± 0.004	0.956 ± 0.004

Table 5

Overall performance for fake news detection on GossipCop.

Methods	ACC	Pre	Rec	F1
TextCNN	0.487 ± 0.008	0.332 ± 0.080	0.545 ± 0.013	0.406 ± 0.059
HAN	0.502 ± 0.009	0.494 ± 0.038	0.544 ± 0.007	0.517 ± 0.022
BERT	0.776 ± 0.007	0.803 ± 0.058	0.769 ± 0.022	0.792 ± 0.015
TextGCN	0.675 ± 0.111	0.590 ± 0.213	0.656 ± 0.130	0.598 ± 0.201
HyperGAT	0.781 ± 0.004	0.781 ± 0.006	0.777 ± 0.004	0.778 ± 0.004
DualEmo	0.816 ± 0.008	0.817 ± 0.009	0.814 ± 0.010	0.814 ± 0.009
HGFND	0.976 ± 0.003	0.973 ± 0.002	0.982 ± 0.007	0.977 ± 0.003
Llama 2	0.975 ± 0.008	0.974 ± 0.007	0.970 ± 0.010	0.972 ± 0.009
SMAN	0.954 ± 0.011	0.951 ± 0.013	0.949 ± 0.013	0.950 ± 0.012
Hy-DeFake	0.979 ± 0.007	0.980 ± 0.007	0.978 ± 0.007	0.979 ± 0.007

social networks as well. However, it still relies on vanilla GNNs to learn hypergraph representations and is limited in capturing sufficient high-order relations. Moreover, HGFND requires numerous types of hyperedges when extracting hypergraphs, increasing the difficulty of dataset construction and impacting real-world practicality. The overall results of SMAN are also notable, as this method takes into accounts the influence of both user and publisher credibility on news authenticity. However, since it does not capture higher-order complicated relations, its performance is not as effective as our model. DualEmo's performance verifies that considering social context *e.g.*, dual emotion between publishers and users, is valuable for distinguishing between real and fake news.

HyperGAT is the other baseline method that uses the concept of hypergraph, but it builds hypergraphs of words in a piece of news, which leads to worse performance than our method due to ignoring user information. Moreover, the construction of hypergraphs makes HyperGAT require the length of text. A hypergraph cannot be built if the text is too short. However, most COVID-19 news is in the form of short tweets, such as those in the MM-COVID dataset, so HyperGAT does not work on this dataset (mark as “–” in Table 4). The reason why HyperGAT can perform on the other three datasets is that the datasets provide statements for each news. Thus, we incorporate these textual data to serve as supplementary information and make the data meet the requirements of HyperGAT. TextGCN builds graphs at the sentence level, so that the structural information of the sentence can be taken into account. Nevertheless, short news in tweets fails to learn enriched information about the structure among sentences, resulting in mediocre results.

The aforementioned five baseline methods (*i.e.*, HGFND, SMAN, DualEmo, HyperGAT, and TextGCN) take into account structural or social context information, while the rest of baseline methods only consider textual information from news content. Apart from Llama 2, another baseline algorithm that performs stable is BERT, indicating that BERT can learn effective textual features of news. However, BERT’s performance in fake news detection is mediocre and worse than that of Hy-DeFake and other social context-based baselines, which reflects that only learning textual information and ignoring social context like user information leads to limited results. In TextCNN and HAN methods, because they do not learn structural information, are not pre-trained in the large-scale corpus, and do not take social contexts into account, they perform the worst.

Comparing the overall results in Tables 2–5, the methods that consider side information perform better than methods that only use textual information. This demonstrates that social context and structural information are indeed instrumental in detecting fake news. Overall, Hy-DeFake consistently delivers accurate and reliable outcomes for distinguishing fake news in various domains.

5.3. Ablation study

To assess the impact of key components in Hy-DeFake, ablation experiments are conducted to evaluate the performance of the news semantic channel, user credibility channel, and MI-based feature fusion individually. The outcomes of these ablation studies are presented in Fig. 5, where “w Sem” and “w Cre” indicate that the news semantic channel and user credibility channel are solely used, respectively. “Sem + Cre” denotes the model incorporating the news semantic channel and the user credibility channel, while excluding MI-based feature fusion. “Hy-DeFake” is the final model with all components. Fig. 5 illustrates that all evaluation metrics on four databases are significantly increased when the user or news signals are not solely used to train our model.

The ablation studies indicate that the performance of “w Cre” is the poorest. That is because the model solely learns user credibility and neglects the objective of classifying news during training. In this case, user attributes play a limited role and the textual information of news is absent to accomplish accurate detection of fake news. This underscores that Hy-DeFake remains text-dependent; relying solely on user information cannot accomplish effective fake news detection. When relying solely on textual information from news content, the outcomes of “w Sem” are similar to those obtained by text-based language models. This is because they exclusively consider semantic information and disregard the significance of social context in detecting fake news. However, when considering user information simultaneously, the direct concatenation of news embeddings and user embeddings in “Sem + Cre” has a significantly improved effect compared to that of “w Sem”. This implies that incorporating social contexts is crucial for superior performance. By introducing MI-based feature fusion, Hy-DeFake exhibits further improvement. This indicates that imposing mutual information

Table 6

User statistics in PolitiFact dataset.

Statistics	Real news	Fake news
Average number of users in each news	43	46
Average number of following of involved users	2,555	3,468
Average number of followers of involved users	48,150	9,140
Verified Users	1,053	167
Total number of users in all news	14,822	13,371

between news features and user features in a unified representation can effectively pull corresponding news and users closer together in the low-dimensional space while pushing unrelated pairs further apart, ultimately leading to more discriminative learned representations. To conclude, each individual component has a significant contribution to the proposed Hy-DeFake for detecting fake news in online social networks.

5.4. Case study and visualization

To examine the influence of user credibility on news reliability, we choose the PolitiFact dataset to perform a case study. Here we analyze the distinctions between users who are involved in spreading fake news versus real news.

We analyze the disparities in user attributes between real and fake news by presenting key statistics of users involved in news spreading in the PolitiFact dataset, as displayed in Table 6. The average number of users in each piece of news demonstrates that fake news tends to attract more attention compared to real news. For real news, the average number of followers significantly exceeds the average number of following, indicating that the users involved in real news are influential and potentially represent public accounts. However, such discrepancy is less obvious in fake news. Furthermore, “verified users” indicates that the account is authentic and belongs to the person whose name it is listed under, and not to a parody account or troll. Here the number of verified users in real news is nearly 10 times more than that in fake news. This attribute reflecting user credibility is positively correlated with news authenticity. By comparing the total number of users in all news and the average number of users in each news, we observe that the total number of users involved in fake news is less than that in real news, but the average number of users in each fake news surpasses that in real news. It illustrates that users involved in fake news spreading are more active, and some may be malicious users who spread false news in large numbers.

In addition to the above analysis of the relationship between news and users from the perspective of statistics and attributes, we also analyze from the perspective of structure in Fig. 6. In this figure, the orange nodes represent the users involved in fake news while the blue nodes stand for the users who spread real news. The edges link two users who are involved in the same piece of news. We discover that users who spread fake news connect more closely while those involved in real news are evenly distributed. The users spreading fake news form a dense community, as some malicious users deliberately participate in the spread of multiple fake news. Therefore, Fig. 6 demonstrated that exploring structural information in the social context is essential for fake news detection.

Finally, to visually observe the representations learned by Hy-DeFake, we applied t-SNE technology to project the low-dimensional representations of the PolitiFact dataset into a 2D space, as shown in Fig. 7. Comparing the distribution with the original data, Hy-DeFake distinctly projects representations of real and fake news into different regions, demonstrating the effectiveness of our method.

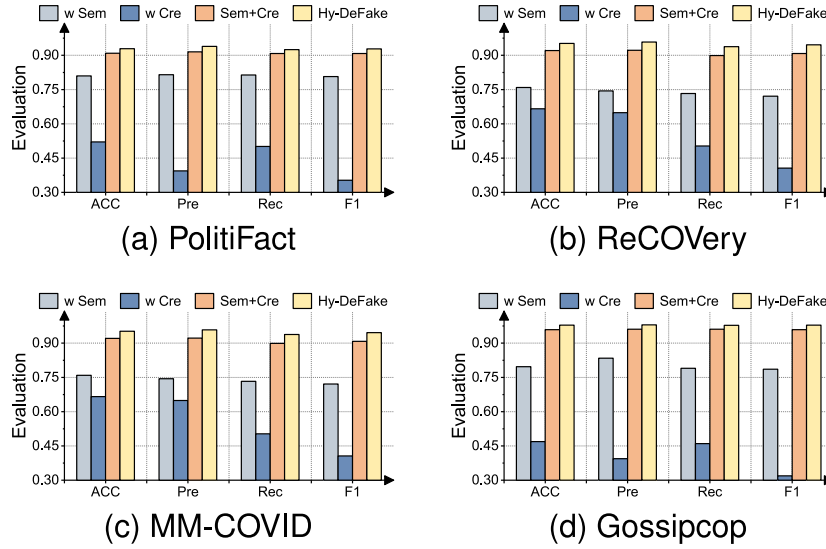


Fig. 5. Experimental results of ablation study on four real-world datasets.



Fig. 6. User relation when spreading news in PolitiFact dataset (Blue nodes: users who spread real news; Orange nodes: users who spread fake news; Edges: connect any two users who participate in the same piece of news).

5.5. Fake news mitigation and early detection

Detecting and mitigating the spread of fake news before it causes harm is crucial. Therefore, we conducted early detection experiments on the PolitiFact dataset to assess Hy-DeFake's applicability in news spreading. Three baseline methods for fake news early detection (i.e., SMAN (Yuan et al., 2020), Propagation2Vec (Silva et al., 2021), and PPC RNN+CNN (Liu & Wu, 2018)) are compared with Hy-DeFake, and the results are presented in Fig. 8. In the experimental settings, we evaluate the ACC performance at intervals of 2 h, 4 h, 8 h, 16 h, 24 h, 36 h, 48 h, and 72 h after the news publication. Notably, Hy-DeFake exhibits satisfactory performance within the first 2 h, achieves above 90 % accuracy after 4 h, and outperforms all baseline models. The experimental results validate the effectiveness of our method in detecting fake news at the early stages of news propagation, owing to its ability to capture higher-order relations in the social context.

5.6. Sensitivity and robustness analysis

This section first analyzes Hy-DeFake's sensitivity to parameter settings, followed by an analysis of its sensitivity to different pooling methods when aggregating user representations. Finally, we evaluate the robustness of Hy-DeFake against adversarial attacks.

5.6.1. Parameter sensitivity

To examine the sensitivity of Hy-DeFake to changes in different parameters, we evaluate the results of different embedding sizes d . As shown in Fig. 9, d is set to be the dimensions of news embeddings and user embeddings. It is crucial to note that Roberta, which we used to update the embeddings of news, requires that the dimension of embeddings be a multiple of the number of attention heads 12. Moreover, the MI loss in updating the embeddings of news and users in the same dimensions. The results demonstrate that Hy-DeFake performs satisfyingly and stably when the embedding dimensions are 1020, 768, and 516.

Through the analysis, we set the dimension of 768 as Hy-DeFake's parameter, because our model performs best in most datasets on this dimension, and the default dimension of textual features learned by the language model is 768. Although on the dimension of 1020, our model can also obtain the optimal results in some metrics, the dimension is too high with inefficiency and the result has little difference with the dimension of 768. Hy-DeFake performs slightly worse on the 516 dimensions than on the 768 dimensions. When the dimension decreased from 252 to 60, the performance of our model gradually deteriorated. This decline is due to the gradual loss of real latent textual and structural information. Overall, our method maintains satisfactory results across most parameter settings.

5.6.2. Pooling sensitivity of user embedding aggregation

Hy-DeFake needs to aggregate the credibility embeddings of users to form the representative embeddings of users that align with the news embeddings for further integration. Table 7 compares the sensitivity of Hy-DeFake to different pooling methods. The results demonstrate that Hy-DeFake achieves satisfactory performance across all four pooling methods, with the element-wise mean pooling yielding the best results.

5.6.3. Robustness against adversarial attacks

News in social networks often faces various adversarial attacks and noise. To verify the robustness of our model against these attacks and noise, we designed two different levels of attacks: text attacks and structural attacks. We simulate text attacks by disrupting 30% of the textual content of news. Structural attacks are modeled by randomly altering the connections of 30% of the nodes to hyperedges in the hypergraph, thereby damaging the true structure of user participation in news dissemination.

Table 8 presents the results of Hy-DeFake under adversarial attacks on both text and structure. It can be observed that textual attacks

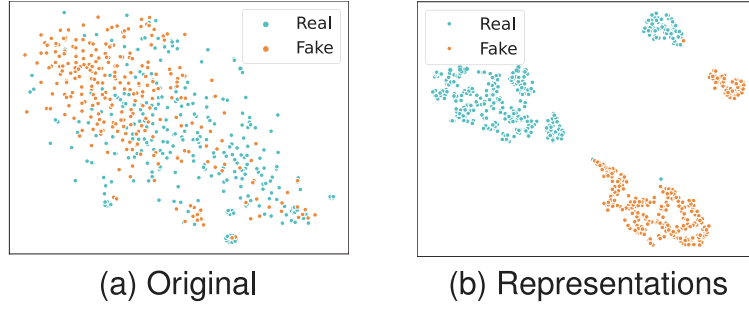


Fig. 7. t-SNE visualization of PolitiFact dataset.

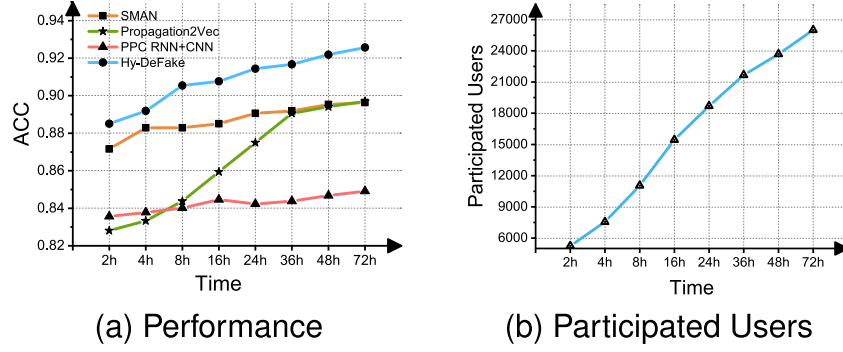
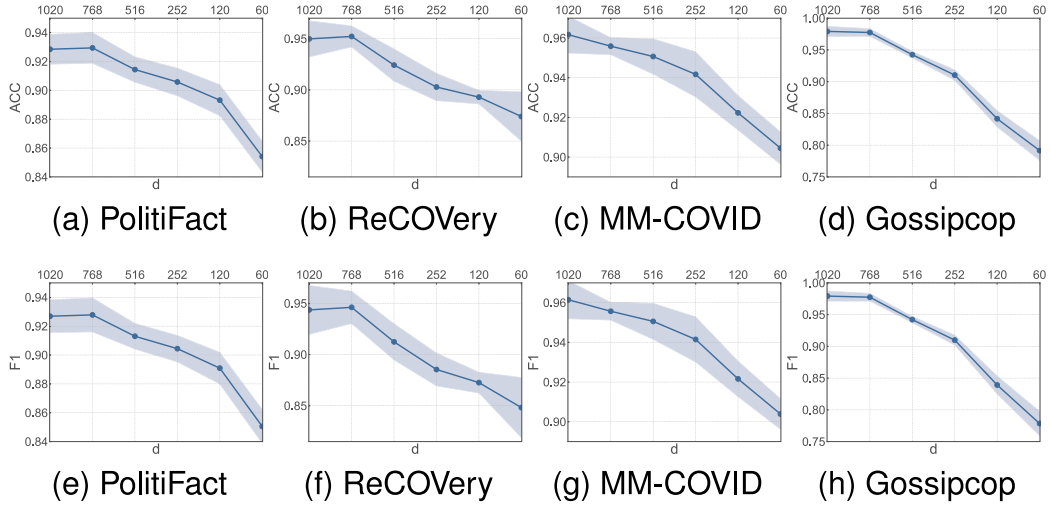
Fig. 8. Results of fake news early detection during news spreading on PolitiFact dataset. (a) **Performance** illustrates the variation of models' accuracy over time, while (b) **Participated Users** depicts the evolution of users engaged in the dissemination of news over time.

Fig. 9. Parameter Analysis.

Table 7
Different pooling for representative user embedding aggregation.

Dataset	Metric	Max	Sum	Attention	Mean
PolitiFact	ACC	0.91 ± 0.01	0.90 ± 0.01	0.92 ± 0.01	0.93 ± 0.01
	F1	0.91 ± 0.01	0.90 ± 0.01	0.92 ± 0.01	0.93 ± 0.01
ReCOvery	ACC	0.95 ± 0.01	0.94 ± 0.01	0.96 ± 0.01	0.95 ± 0.01
	F1	0.94 ± 0.01	0.93 ± 0.01	0.95 ± 0.01	0.95 ± 0.01
MM-COVID	ACC	0.93 ± 0.01	0.93 ± 0.01	0.95 ± 0.01	0.96 ± 0.01
	F1	0.93 ± 0.01	0.93 ± 0.01	0.95 ± 0.01	0.96 ± 0.01
GossipCop	ACC	0.96 ± 0.02	0.96 ± 0.02	0.97 ± 0.01	0.98 ± 0.01
	F1	0.96 ± 0.02	0.96 ± 0.02	0.97 ± 0.01	0.98 ± 0.01

Table 8
Adversarial attack on text and structure.

Dataset	Metric	No attack	Text attack	Structural attack
PolitiFact	ACC	0.929 ± 0.009	0.930 ± 0.01	0.911 ± 0.016
	F1	0.928 ± 0.011	0.928 ± 0.011	0.909 ± 0.017
ReCOvery	ACC	0.952 ± 0.009	0.955 ± 0.007	0.938 ± 0.009
	F1	0.946 ± 0.014	0.950 ± 0.011	0.929 ± 0.012
MM-COVID	ACC	0.956 ± 0.004	0.953 ± 0.004	0.926 ± 0.026
	F1	0.956 ± 0.004	0.954 ± 0.004	0.925 ± 0.026
GossipCop	ACC	0.979 ± 0.007	0.977 ± 0.005	0.971 ± 0.006
	F1	0.979 ± 0.007	0.976 ± 0.006	0.969 ± 0.006

Table 9

Average training time on each epoch.

Datasets	PolitiFact	ReCOVery	MM-COVID	Gossipcop
AV. Time	2.42 s	6.27 s	1.92 s	86.38 s

have no significant impact on our model. This is because the higher-order relations learned by hypergraphs can effectively mitigate the noise in semantic information. Under structural attacks, there is a slight performance drop, highlighting the importance of capturing higher-order structural information. Overall, Hy-DeFake demonstrates strong robustness in the face of both types of attacks.

5.7. Time efficiency

To evaluate the execution time of Hy-DeFake, we record its training time on four datasets. Due to the variations in input sizes (e.g., the hypergraph describing users spreading the news in Hy-DeFake is not considered by text-based baselines), comparing the runtime directly with other baseline methods is unfair. Table 9 presents the average training time per epoch for Hy-DeFake on each dataset. The results demonstrate that Hy-DeFake achieves efficient and effective training, with superior performance on four datasets. The training time of Gossipcop is the longest, since the large number of users in this dataset. Overall, Hy-DeFake learns the most informative embeddings in an acceptable time and achieves superior and more stable results than baseline methods.

5.8. Discussion and future work

In this section, we delve deeper into the impact of users in fake news detection. Section of 5.4 (Case Study and Visualization) reveals distinct differences in both attributes and high-order structures between users who engage with real news versus fake news. The discovery of this phenomenon presents a potential opportunity to identify fake news without text analysis, by leveraging high-order relations among users in social networks, i.e., integrating user attributes with their structural connections. Exploring this opportunity is promising for domain-specific news detection, because investigating this news, such as healthcare news, requires domain knowledge and expertise. However, it is challenging for the public to acquire such knowledge. Thus, in our future work, we will explore whether it is possible to address this issue by capturing high-order structural information without analyzing the textual content of news.

Additionally, we recognize the importance of addressing bias, particularly in detecting fake news from diverse outlets and the potential bias introduced by fact-checked news sources. We will further investigate methods to reduce such bias and ensure our model remains impartial across different political and ideological contexts. Furthermore, we will explore strategies to address both label noise and dataset imbalance, as this is essential for enhancing the overall performance and robustness of our fake news detection model. Since news is generally disseminated on large-scale social networks, the efficient and early detection of fake news in large-scale datasets is an important area for future research. Lastly, Exploring user stance detection and fake news detection as a multi-task learning problem is also worthwhile, as user stances provide response signals to enhance fake news identification.

6. Conclusion

In this work, our task is to detect fake news in online social networks, and we argue that existing methods face two challenges in doing so. To address these challenges, we explore the high-order correlation between news and users. Firstly we construct an attributed hypergraph to capture the intricate relationships between news and users in online social networks. Subsequently, we propose Hy-DeFake, a hypergraph

neural network-based method for detecting fake news. Hy-DeFake not only learns semantic embeddings of news content and credibility embeddings of users, but also incorporates high-order correlations between news and users. By integrating these embeddings, Hy-DeFake can provide informative embeddings for real and fake news classification. Extensive experiments on four real-world datasets demonstrate the superior performance of Hy-DeFake, indicating that it learns distinctive embeddings that contain rich semantic and credibility information as well as high-order correlations. Furthermore, our findings verify a positive correlation between news authority and user credibility. Users who spread fake news exhibit more intensive interaction compared to those who spread real news, resulting in the formation of a denser community.

CRedit authorship contribution statement

Xing Su: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Data curation, Conceptualization. **Jian Yang:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Jia Wu:** Writing – review & editing, Supervision, Project administration. **Zitai Qiu:** Visualization, Validation, Software.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jia Wu, Jian Yang reports financial support was provided by Australian Research Council. Jia Wu reports a relationship with Australian Research Council that includes: funding grants. Jian Yang reports a relationship with Australian Research Council that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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